12.892

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

The eigenvalues of the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$$

are λ_1, λ_2 and λ_3. Find the value of

$$\lambda_1\lambda_2\lambda_3(\lambda_1 + \lambda_2 + \lambda_3).$$

Solution:
We know that for any square matrix:

$$\lambda_1\lambda_2\lambda_3 = \det \mathbf{A}, \qquad \lambda_1 + \lambda_2 + \lambda_3 = \text{trace}(\mathbf{A}). \tag{1}$$

$$\text{trace}(\mathbf{A}) = 3 + 5 + 3 = 11. \tag{2}$$

$$\det \mathbf{A} = 3(15 - 1) - (-1)(-3 + 1) + (1 - 5) \tag{3}$$

$$= 3 \cdot 14 - 2 - 4 \tag{4}$$

$$= 36. \tag{5}$$

Hence

$$\lambda_1\lambda_2\lambda_3(\lambda_1 + \lambda_2 + \lambda_3) = \det \mathbf{A} \cdot \text{trace}(\mathbf{A}) = 36 \cdot 11 = 396. \tag{6}$$